3.3 Vectors in the Plane

Pre-Calculus 3.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- write the component form of a vector and find its magnitude.
- add, subtract, and multiply vectors by scalars, and find unit vectors.
- use direction angles to write vectors and solve real-life problems.

Vocabulary · fill in the missing word

Vector — a quantity with _____ and direction, shown as a directed line segment PQ (initial point P, terminal point Q)

Component form — $v = \langle$ _____, $v_2\rangle = \langle q_1 - p_1, q_2 - p_2\rangle$

Magnitude — $\|v\| =$ _____

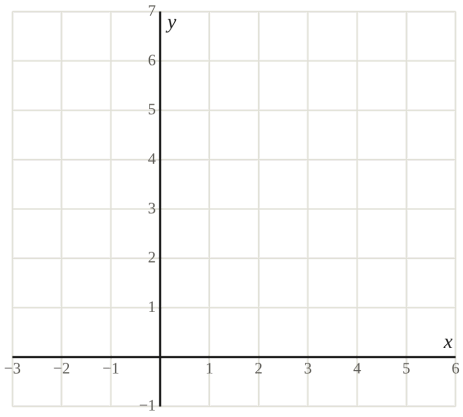
Unit vector — a vector with _____ 1: $v/\|v\|$

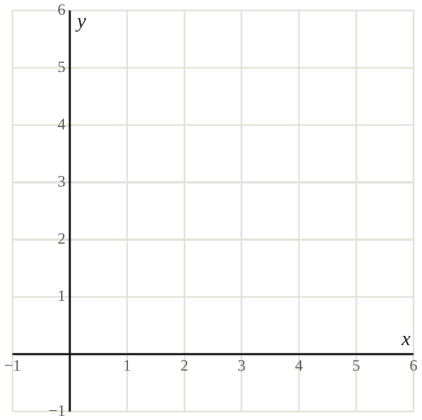
Standard unit vectors — $i =$ _____, $0\rangle$ and $j = \langle 0, 1\rangle$

Direction angle — the angle θ from the _____ x-axis: $v = \|v\|\langle\cos\theta, \sin\theta\rangle$

Part 1 · Arrows on a grid

Grid 1: draw an arrow from (1, 1) to (4, 5) and another from (−2, 0) to (1, 4). Grid 2: draw $u = \langle 3, 1\rangle$ starting at the origin, then $v = \langle 1, 3\rangle$ starting at the tip of u.





Arrow	Right	Up	Length
(1, 1) to (4, 5)			
(−2, 0) to (1, 4)			

From the drawing, $u + v = \langle$_____, _____$\rangle$. From the components: _____

Part 2 · Notes: fill in the blanks

The component form of the vector from $P(p_1, p_2)$ to $Q(q_1, q_2)$ is _____.

Magnitude: $\|v\| =$ _____

$u + v =$ _____ $ku =$ _____

A unit vector in the direction of v is _____.

Direction angle: $v = \|v\|\langle$_____, _____$\rangle$

Part 3 · Examples (work them with the class)

Example 1

Find the component form and magnitude of the vector with initial point P(4, −7) and terminal point Q(−1, 5).

Example 2

Let $u = \langle -2, 5 \rangle$ and $v = \langle 3, 4 \rangle$. Find each vector.

a. $2v$ b. $u + v$ c. $v - u$ d. $2u + 3v$

Example 3

Find a unit vector in the direction of $v = \langle -2, 5 \rangle$. Verify that it has magnitude 1.

Example 4

Let u be the vector with initial point (−1, 3) and terminal point (2, −5). Write u as a linear combination of i and j.

Example 5

Find the magnitude and direction angle of each vector.

a. $u = 3i + 3j$ b. $v = 3i - 4j$

Example 6

Find the component form of v with $\|v\| = 6$ and direction angle 150°.

Example 7

A ball is thrown at 70 feet per second at an angle of 35° above the horizontal. Find the horizontal and vertical components of its velocity.

Part 4 · Partner practice

Partner 1

Let $u = \langle 4, -1 \rangle$ and $v = \langle -2, 3 \rangle$. Find $3u - v$ and its magnitude.

Partner 2

Find the magnitude and direction angle of $v = -4i + 4j$.